



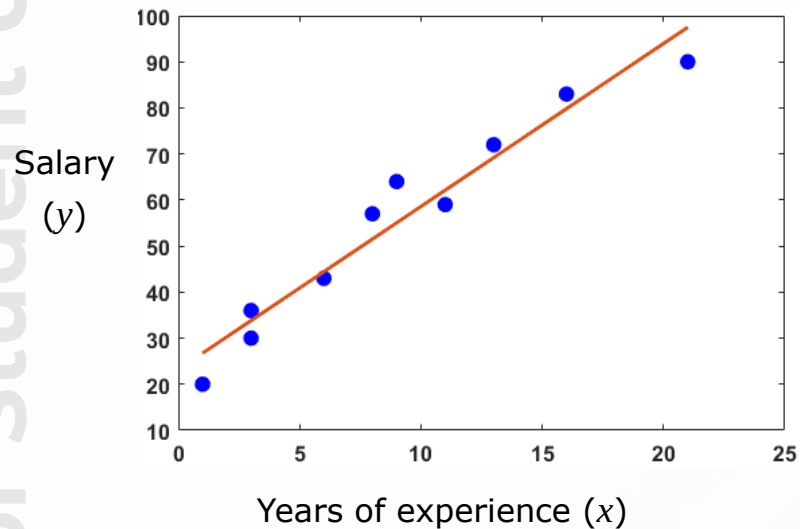
Supervised Machine Learning: Regression Methods

Numeric Prediction (Regression)

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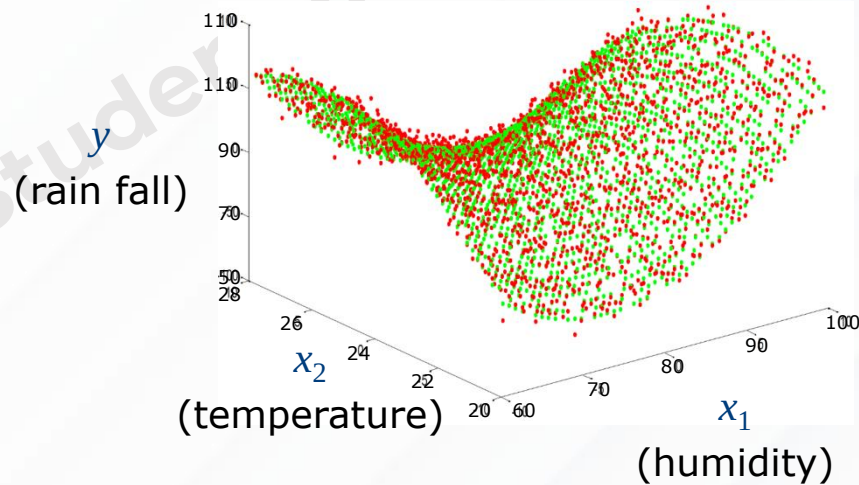
- **Numeric prediction:** Task of predicting continuous values for given input
- Example:

- Predicting **salary** given its **year of experience**



- Fitting a **line/curve**

- Predicting **amount of rain fall** given the **temperature** and **humidity** in the atmosphere



$$\mathbf{x} = [x_1, x_2]^T$$

- Fitting a **surface**

Numeric Prediction (Regression)

- **Regression analysis** is used to model the relationship between **one or more independent** (input) variable and a **dependent** (output) variable
 - Dependent variable is always **continuous valued** or **ordered valued**
 - **Example:** Predicting amount of rain fall given the temperature and humidity in the atmosphere
 - **Dependent variable:** amount of rain fall
 - **Independent variable(s):** temperature, humidity
- The values of independent variables are known
- The dependent variable is what we want to predict
- Regression analysis can be viewed as mapping function:



- Single independent variable (x)
- Single dependent variable (y)



- Multiple independent variable ($\mathbf{x} \in \mathbb{R}^d$)
- Single dependent variable (y)

Numeric Prediction (Regression)

- Regression is a **two-step** process
 - **Step1:** Building a regression model
 - **Learning from data** (training phase)
 - **Supervised learning:**
 - In supervised learning, each example is a **pair** consisting of an **input example** (independent variables) and a **desired output value** (dependent variable)
 - Regression model is built by analyzing or learning from a training data set made up of one or more input variables and their output variable
 - Learn the mapping function (association) between input and output

$$y_n = f(\mathbf{x}_n)$$

- $\mathbf{x}_n$ is the n^{th} training example and y_n is the associated class label
- **Step2:** Using regression model for prediction
 - Testing phase - Predicting output variable
- **Accuracy of a predictor:**
 - How well a given predictor can predict for new values
- Target of learning techniques: **Good generalization** ability
 - Performance of the learnt model on the unseen data (data not seen during training) should be good

Illustration of Training Set: Salary Prediction

Years of experience (x)	Salary (in Rs 1000) (y)
3	30
8	57
9	64
13	72
3	36
6	43
11	59
21	90
1	20
16	83

- Independent variable: Years of experience (x)
- Dependent variable: Salary (y)

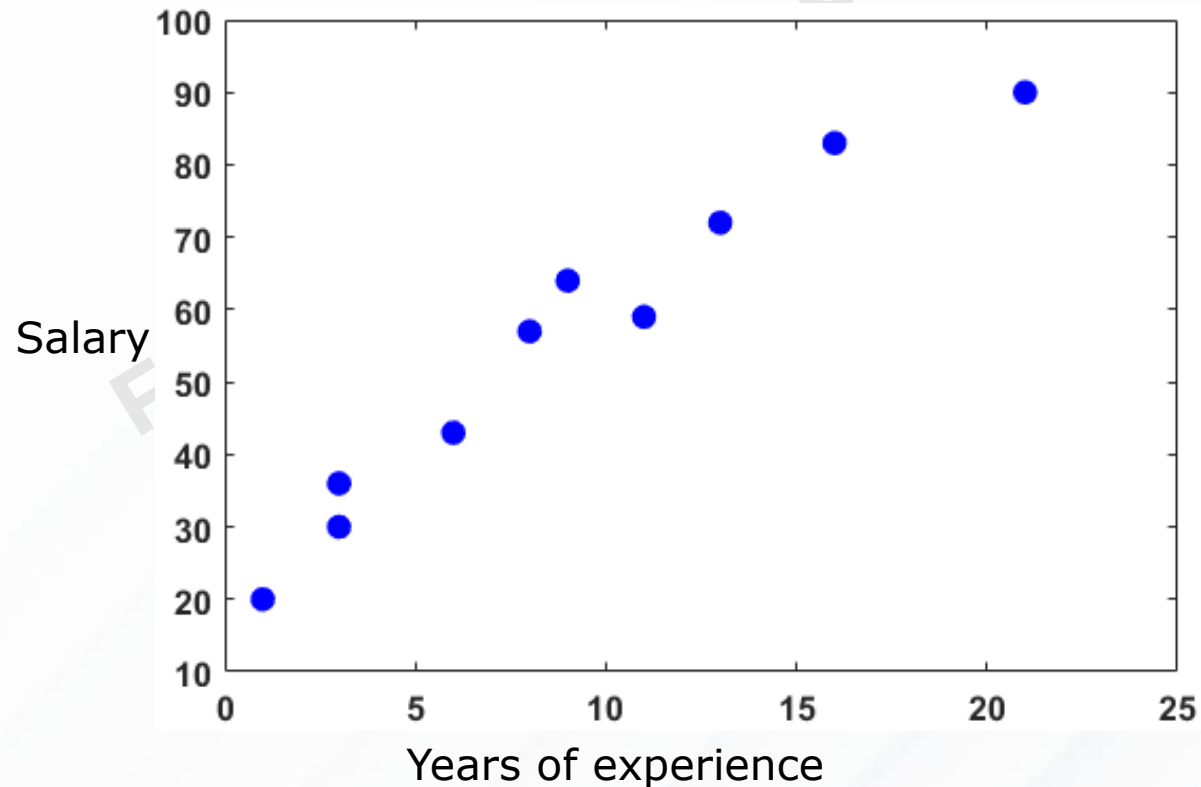


Illustration of Training Set: Temperature Prediction

Humidity (x_1)	Pressure (x_2)	Temp (y)
82.19	1036.35	25.47
83.15	1037.60	26.19
85.34	1037.89	25.17
87.69	1036.86	24.30
87.65	1027.83	24.07
95.95	1006.92	21.21
96.17	1006.57	23.49
98.59	1009.42	21.79
88.33	991.65	25.09
90.43	1009.66	25.39
94.54	1009.27	23.89
99.00	1009.80	22.51
98.00	1009.90	22.90
99.00	996.29	21.72
98.97	800.00	23.18

- Independent variables: Humidity, Pressure

$$\mathbf{x} = [x_1 \ x_2]^T$$

- Dependent variable, y :
Temperature (Temp)

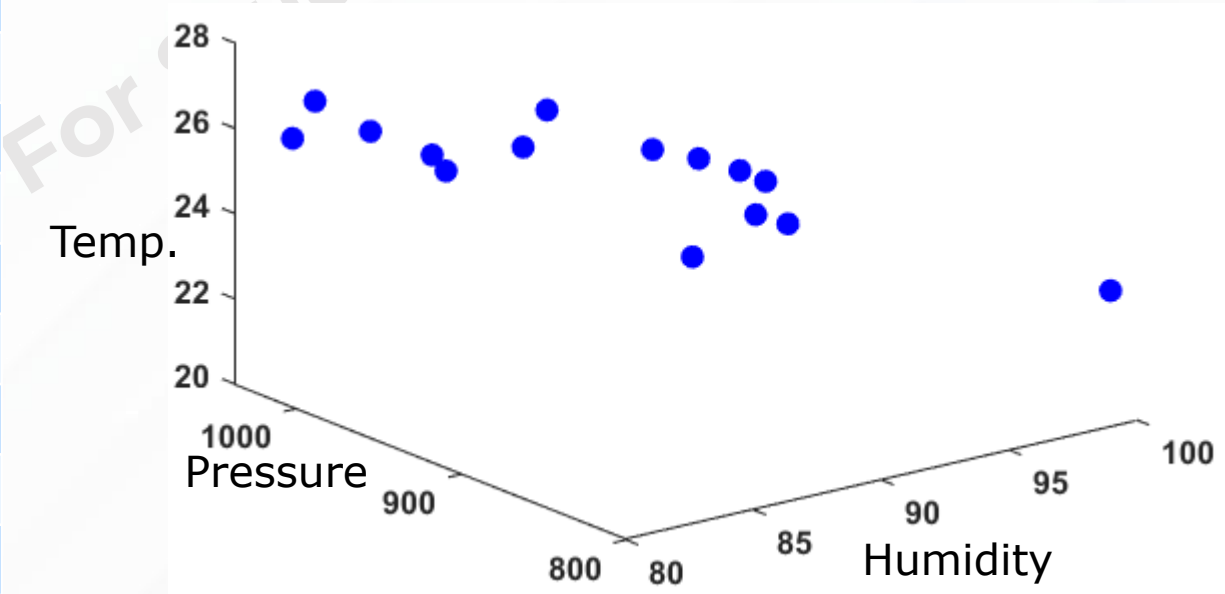


Illustration of Training Set: Wine Quality Prediction [1]

- Independent variables: $\mathbf{x} = [x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}]^T$
- Dependent variable, y : **Quality**

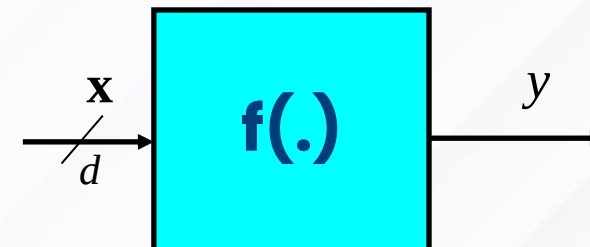
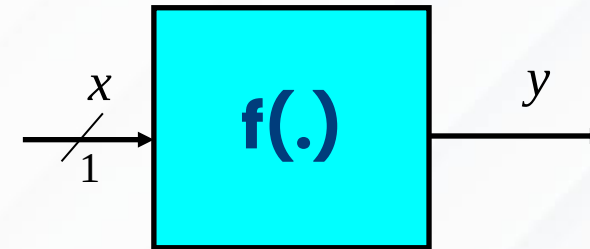
Fixed Acidity (x_1)	Volatile Acidity (x_2)	Citric acid (x_3)	Residual Sugar (x_4)	Chlorides (x_5)	Free SO ₂ (x_6)	Total SO ₂ (x_7)	Density (x_8)	pH (x_8)	Sulphates (x_9)	Alcohol (x_{10})	Quality (y)
7.4	0.7	0	1.9	0.076	11	34	0.9978	3.51	0.56	9.4	5.42
7.8	0.88	0	2.6	0.098	25	67	0.9968	3.2	0.68	9.8	5.57
7.8	0.76	0.04	2.3	0.092	15	54	0.997	3.26	0.65	9.8	5.17
11.2	0.28	0.56	1.9	0.075	17	60	0.998	3.16	0.58	9.8	6.65
7.4	0.7	0	1.9	0.076	11	34	0.9978	3.51	0.56	9.4	5.68
7.4	0.66	0	1.8	0.075	13	40	0.9978	3.51	0.56	9.4	5.63
7.9	0.6	0.06	1.6	0.069	15	59	0.9964	3.3	0.46	9.4	5.32
7.3	0.65	0	1.2	0.065	15	21	0.9946	3.39	0.47	10	7.16
7.8	0.58	0.02	2	0.073	9	18	0.9968	3.36	0.57	9.5	7.2
7.5	0.5	0.36	6.1	0.071	17	102	0.9978	3.35	0.8	10.5	5.18

[1] P. Cortez, A. Cerdeira, F. Almeida, T. Matos and J. Reis. "Modeling wine preferences by data mining from physicochemical properties," In Decision Support Systems, Elsevier, vol. 47, issue 4, pp. 547-553, 2009..

Linear Regression

Linear Regression

- **Linear approach** to model the relationship between a scalar response, (y) (or **dependent variable**) and one or more predictor variables, $(x$ or $\mathbf{x})$ (or **independent variables**)
- The output is going to be the **linear function** of input (one or more independent variables)
- **Simple linear regression (straight-line regression):**
 - Single independent variable (x)
 - Single dependent variable (y)
 - **Fitting a straight-line**
- **Multiple linear regression:**
 - Two or more independent variable $(\mathbf{x})$
 - Single dependent variable (y)
 - **Fitting a hyperplane (linear surface)**



Straight-Line (Simple Linear) Regression

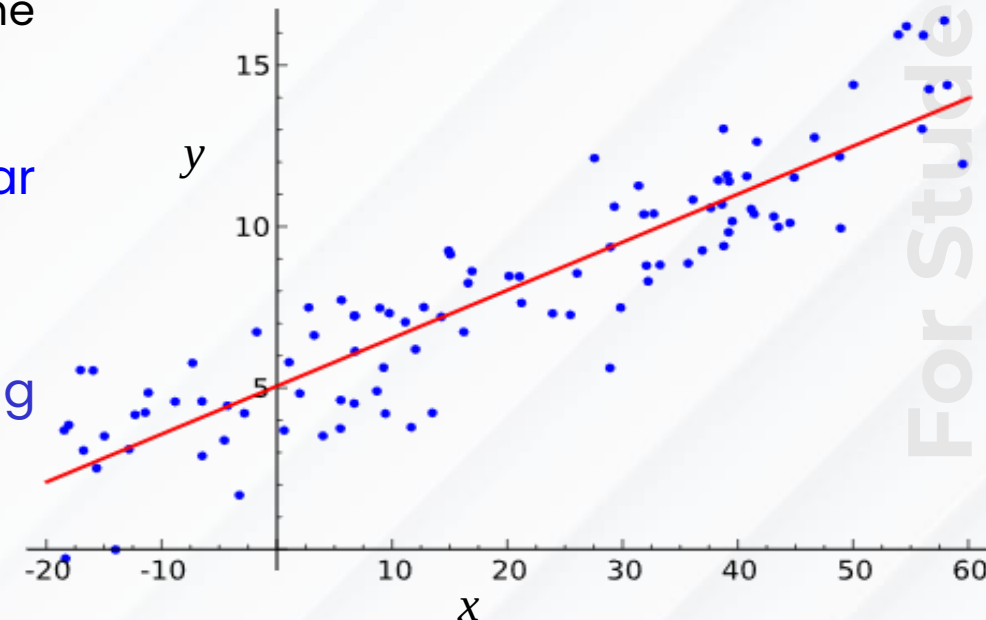
- Given:- **Training data:** $\mathcal{D} = \{x_n, y_n\}_{n=1}^N$, $x_n \in \mathbb{R}^1$ and $y_n \in \mathbb{R}^1$
 - x_n : n^{th} input example (independent variable)
 - y_n : Dependent variable (output) corresponding to n^{th} independent variable
- Example:** Predicting the salary given the year of experience

Years of experience (x)	Salary (in Rs 1000) (y)
3	30
8	57
9	64
13	72
3	36
6	43
11	59
21	90
1	20
16	83

- Independent variable:
 - Years of experience
- Dependent variable:
 - Salary

Straight-Line (Simple Linear) Regression

- Given:- **Training data:** $\mathcal{D} = \{x_n, y_n\}_{n=1}^N$, $x_n \in \mathbb{R}^1$ and $y_n \in \mathbb{R}^1$
 - x_n : n^{th} input example (independent variable)
 - y_n : Dependent variable (output) corresponding to n^{th} independent variable
- Function governing the relationship between input and output:**
$$y_n = f(x_n, w, w_0) = w x_n + w_0$$
 - The coefficients w_0 and w are parameters of straight-line (regression coefficients) - **Unknown**
- Function $f(x_n, w, w_0)$ is a linear function of x_n and it is a linear function of coefficients w and w_0
 - **Linear model for regression**
- The values for the coefficients will be determined by fitting the linear function (straight-line) to the training data



Straight-Line (Simple Linear) Regression: Training Phase

- Given:- **Training data:** $\mathcal{D} = \{x_n, y_n\}_{n=1}^N$, $x_n \in \mathbb{R}^1$ and $y_n \in \mathbb{R}^1$
- **Method of least squares:** Minimizes the sum of the squared error between
 - all the actual data (y_n) i.e. actual dependent variable and
 - the estimate of line (predicted dependent variable ($\hat{y}_n$)) i.e. the function $f(x_n, w, w_0)$, in the training set for any given value of w and w_0

$$\hat{y}_n = f(x_n, w, w_0) = w x_n + w_0$$

$$(\hat{y}_n - y_n)^2 \quad \forall n = 1, 2, \dots, N$$

Straight-Line (Simple Linear) Regression: Training Phase

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$$\hat{y}_n = f(x_n, w, w_0) = w x_n + w_0$$

$$E(w, w_0) = \frac{1}{2} \sum_{n=1}^N (\hat{y}_n - y_n)^2$$

Straight-Line (Simple Linear) Regression: Training Phase

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$$\hat{y}_n = f(x_n, w, w_0) = w x_n + w_0$$

$$E(w, w_0) = \frac{1}{2} \sum_{n=1}^N (f(x_n, w, w_0) - y_n)^2$$

Straight-Line (Simple Linear) Regression: Training Phase

- Given:- **Training data:** $\mathcal{D} = \{x_n, y_n\}_{n=1}^N$, $x_n \in \mathbb{R}^1$ and $y_n \in \mathbb{R}^1$
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$$\hat{y}_n = f(x_n, w, w_0) = w x_n + w_0$$

$$\underset{w, w_0}{\text{minimize}} E(w, w_0) = \frac{1}{2} \sum_{n=1}^N (f(x_n, w, w_0) - y_n)^2$$

- Minimize the error such that the coefficients w_0 and w represent the parameter of line that best fit the training data

Straight-Line (Simple Linear) Regression: Training Phase

- Given:- **Training data:** $\mathcal{D} = \{x_n, y_n\}_{n=1}^N$, $x_n \in \mathbb{R}^1$ and $y_n \in \mathbb{R}^1$
- **Method of least squares:** Minimizes the sum of the squared error between
 - all the actual data (y_n) i.e. actual dependent variable and
 - the estimate of line (predicted dependent variable ($\hat{y}_n$)) i.e. the function $f(x_n, w, w_0)$, in the training set for any given value of w and w_0

$$\hat{y}_n = f(x_n, w, w_0) = w x_n + w_0$$

$$\underset{w, w_0}{\text{minimize}} E(w, w_0) = \frac{1}{2} \sum_{n=1}^N (f(x_n, w, w_0) - y_n)^2$$

- The derivatives of error function with respect to the coefficients will be linear in the elements of w and w_0
- Hence the minimization of the error function has unique solution and found in closed form

Straight-Line (Simple Linear) Regression: Training Phase

- Cost function for optimization:

$$E(w, w_0) = \frac{1}{2} \sum_{n=1}^N (f(x_n, w, w_0) - y_n)^2$$

- Conditions for optimality: $\frac{\partial E(w, w_0)}{\partial w} = 0$ $\frac{\partial E(w, w_0)}{\partial w_0} = 0$

$$\frac{\partial \frac{1}{2} \sum_{n=1}^N (w x_n + w_0 - y_n)^2}{\partial w} = 0 \quad \frac{\partial \frac{1}{2} \sum_{n=1}^N (w x_n + w_0 - y_n)^2}{\partial w_0} = 0$$

- Solving this give optimal $\hat{w}$ and $\hat{w}_0$ as

$$\hat{w} = \frac{\sum_{n=1}^N (x_n - \mu_x)(y_n - \mu_y)}{\sum_{n=1}^N (x_n - \mu_x)^2}$$

$$\hat{w}_0 = \mu_y - \hat{w} \mu_x$$

- μ_x : sample mean of independent variable x
- μ_y : sample mean of dependent variable y

Straight-Line (Simple Linear)

Regression: Testing Phase

- For any test example x , the predicted value is given by:

$$\hat{y} = f(x, \hat{w}, \hat{w}_0) = \hat{w} x + \hat{w}_0$$

- For any $\hat{w}$ and $\hat{w}_0$ are the optimal parameters of the line learnt during training

Evaluation Metrics for Regression:

Squared Error and Mean Squared Error

- The prediction accuracy is measured in terms of squared error:

$$E = (\hat{y} - y)^2$$

- y : actual value
- $\hat{y}$: predicted value
- Let N_t be the total number of test samples
- The prediction accuracy of regression model is measured in terms of **root mean squared error (RMSE)**:

$$E_{\text{RMSE}} = \sqrt{\frac{1}{N_t} \sum_{n=1}^{N_t} (\hat{y}_n - y_n)^2}$$

- RMSE expressed in % as:
$$E_{\text{RMSE}} (\%) = \frac{\sqrt{\frac{1}{N_t} \sum_{n=1}^{N_t} (\hat{y}_n - y_n)^2}}{\frac{1}{N_t} \sum_{n=1}^{N_t} y_n} \times 100$$

Illustration of Simple Linear Regression: Salary Prediction – Training

Years of experience (x)	Salary (in Rs 1000) (y)
3	30
8	57
9	64
13	72
3	36
6	43
11	59
21	90
1	20
16	83

$$\hat{W} = \frac{\sum_{n=1}^N (x_n - \mu_x)(y_n - \mu_y)}{\sum_{n=1}^N (x_n - \mu_x)^2}$$

$$\hat{W}_0 = \mu_y - \hat{W}\mu_x$$

- μ_x : 9.1
- μ_y : 55.4
- $\hat{W}$: 3.54
- $\hat{W}_0$: 23.21

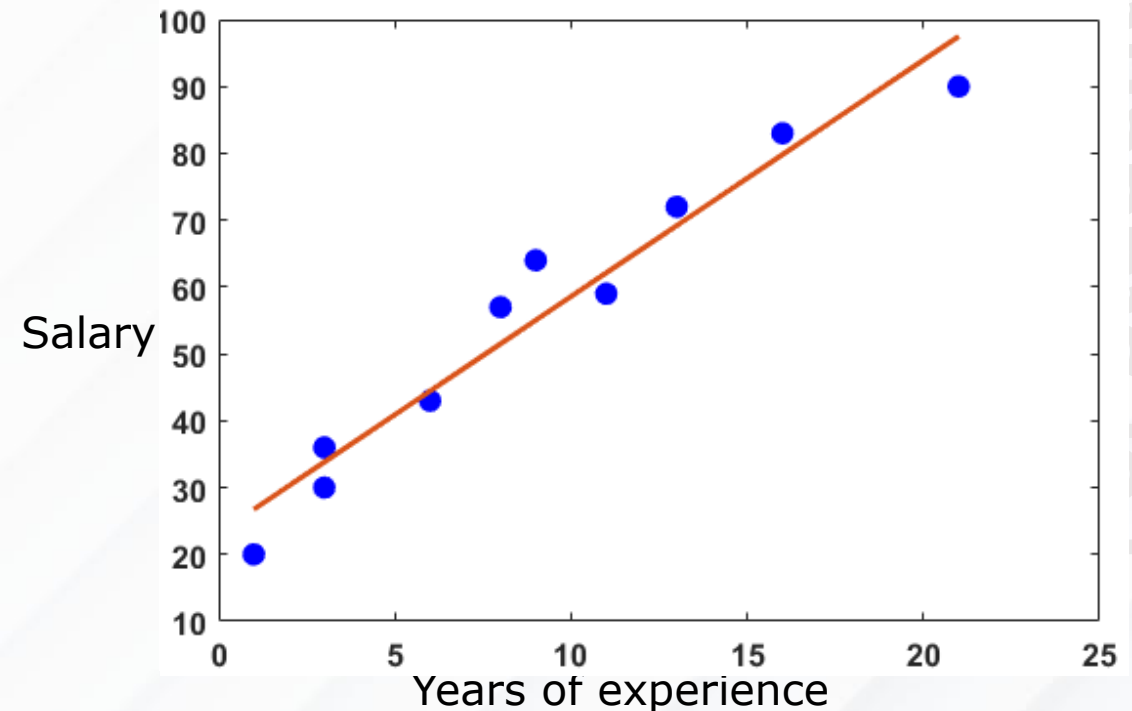


Illustration of Simple Linear Regression: Salary Prediction – Test

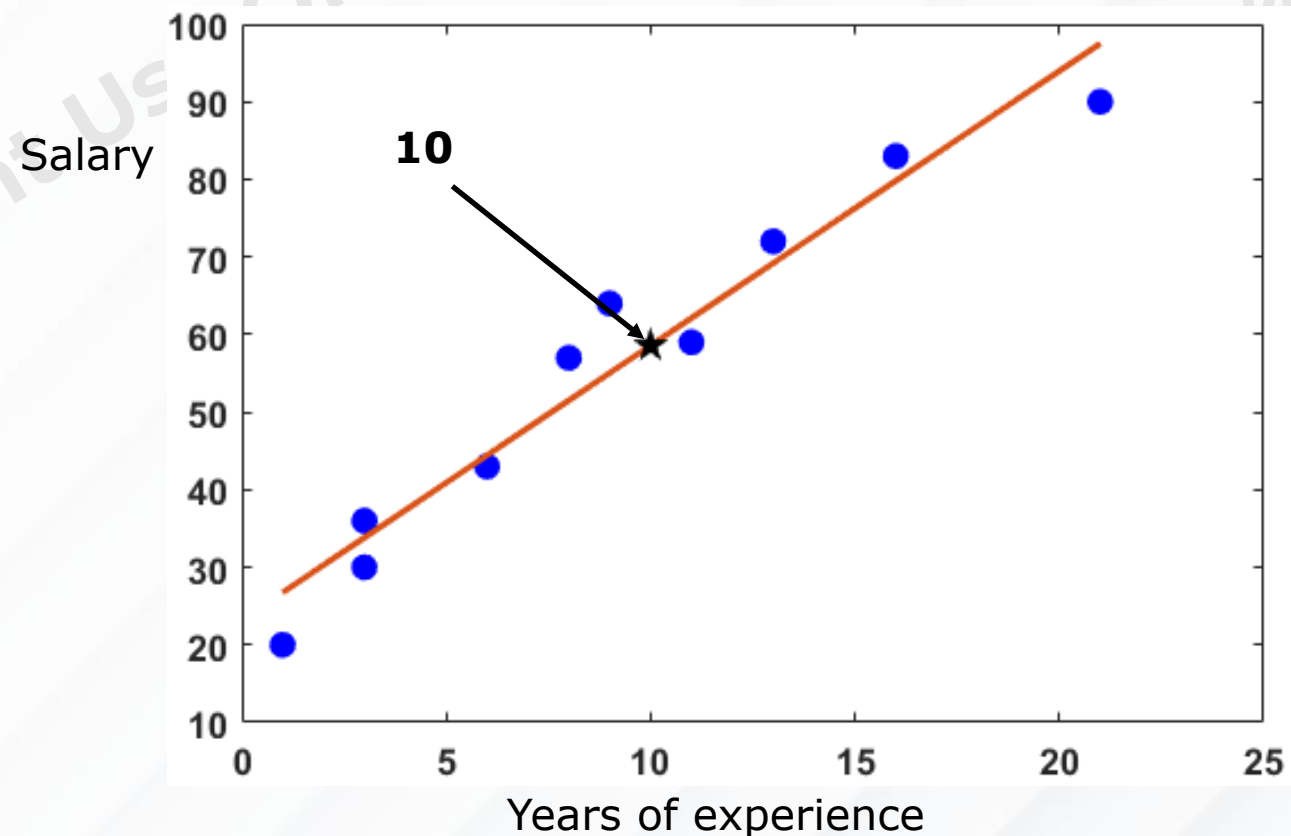
- $\hat{w}$: 3.54
- $\hat{w}_0$: 23.21

Years of experience (x)	Salary (in Rs 1000) (y)
10	-

$$\hat{y} = \hat{w} x + \hat{w}_0$$

$$\hat{y} = 3.54 * 10 + 23.21$$

- Predicted salary, $\hat{y}$: 58.584
- Actual salary, y : 58.000
- Squared error, $E = (\hat{y} - y)^2$: 0.34

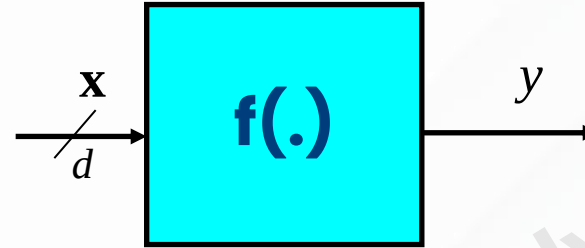


Multiple Linear Regression

Multiple Linear Regression

- **Multiple linear regression:**

- Two or more independent variable ($\mathbf{x}$)
- Single dependent variable (y)



- Given:- **Training data:** $\mathcal{D} = \{\mathbf{x}_n, y_n\}_{n=1}^N$, $\mathbf{x}_n \in \mathbb{R}^d$ and $y_n \in \mathbb{R}^1$

- d : dimension of input example (number of independent variables)
- $\mathbf{x}_n$: n^{th} input example (d independent variables) $\mathbf{x}_n = [x_{n1}, x_{n2}, \dots, x_{nd}]^T$
- y_n : Dependent variable (output) corresponding to n^{th} input example

- **Function governing the relationship between input and output:**

$$y_n = f(\mathbf{x}_n, \mathbf{w}) = w_d x_{nd} + \dots + w_2 x_{n2} + w_1 x_{n1} + w_0 = \sum_{i=0}^d w_i x_{ni} = \mathbf{w}^T \mathbf{x}_n$$

- The coefficients $w_0, w_1, \dots, w_d$ are collectively denoted by the vector $\mathbf{w}$ - **Unknown**
- Function $f(\mathbf{x}_n, \mathbf{w})$ is a linear function of $\mathbf{x}_n$ and it is a linear function of coefficients $\mathbf{w}$
 - **Linear model for regression**

Linear Regression:

Linear Function Approximation

- **Linear function:**

- **2 input variable case** (3-dimensional space): The mapping function is a **plane** specified by

$$y = f(\mathbf{x}, \mathbf{w}) = w_2 x_2 + w_1 x_1 + w_0 = 0$$

$$\text{where } \mathbf{w} = [w_0, w_1, w_2]^T \text{ and } \mathbf{x} = [1, x_1, x_2]^T$$

- **d input variable case** (d+1 – dimensional space): The mapping function is a hyperplane specified by

$$y = f(\mathbf{x}, \mathbf{w}) = w_d x_d + \dots + w_2 x_2 + w_1 x_1 + w_0 = \sum_{i=0}^d w_i x_i = \mathbf{w}^T \mathbf{x} = 0$$

$$\text{where } \mathbf{w} = [w_0, w_1, \dots, w_d]^T \text{ and } \mathbf{x} = [1, x_1, \dots, x_d]^T$$

Multiple Linear Regression: Training Phase

- The values for the coefficients will be determined by fitting the linear function to the training data
- Given:- **Training data:** $\mathcal{D} = \{\mathbf{x}_n, y_n\}_{n=1}^N$, $\mathbf{x}_n \in \mathbb{R}^d$ and $y_n \in \mathbb{R}^1$
- **Method of least squares:** Minimizes the sum of the squared error between
 - all the actual data (y_n) i.e. actual dependent variable and
 - the estimate of line (predicted dependent variable ($\hat{y}_n$)) i.e. the function $f(\mathbf{x}_n, \mathbf{w})$, in the training set for any given value of $\mathbf{w}$

$$\hat{y}_n = f(\mathbf{x}_n, \mathbf{w}) = \mathbf{w}^\top \mathbf{x}_n + w_0 = \sum_{i=0}^d w_i x_i$$

$$\underset{\mathbf{w}}{\text{minimize}} E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (\hat{y}_n - y_n)^2$$

- The error function is a
 - The derivatives of error function with respect to the coefficients will be linear in the elements of $\mathbf{w}$
- Hence the minimization of the error function has unique solution and found in closed form

Multiple Linear Regression: Training Phase

- Cost function for optimization:

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (f(\mathbf{x}_n, \mathbf{w}) - y_n)^2$$

- Conditions for optimality: $\frac{d E(\mathbf{w})}{d \mathbf{w}} = \mathbf{0}$

- Application of optimality conditions gives optimal $\hat{\mathbf{w}}$:

$$\frac{d \frac{1}{2} \sum_{n=1}^N \left(\sum_{i=0}^d w_i x_{ni} - y_n \right)^2}{d \mathbf{w}} = \mathbf{0}$$

$$\frac{d \frac{1}{2} \sum_{n=1}^N (\mathbf{w}^\top \mathbf{x}_n - y_n)^2}{d \mathbf{w}} = \mathbf{0}$$

$$\frac{d E(\mathbf{w})}{d \mathbf{w}} = \begin{bmatrix} \frac{\partial E(\mathbf{w})}{\partial w_0} \\ \frac{\partial E(\mathbf{w})}{\partial w_1} \\ \vdots \\ \frac{\partial E(\mathbf{w})}{\partial w_d} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \mathbf{0}$$

Multiple Linear Regression: Training Phase

- Cost function for optimization:

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (f(\mathbf{x}_n, \mathbf{w}) - y_n)^2$$

- Conditions for optimality: $\frac{d E(\mathbf{w})}{d \mathbf{w}} = \mathbf{0}$

- Application of optimality conditions gives optimal $\hat{\mathbf{w}}$:

$$\frac{d \frac{1}{2} \sum_{n=1}^N (\mathbf{w}^\top \mathbf{x}_n - y_n)^2}{d \mathbf{w}} = \mathbf{0}$$

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

- Assumption: $d < N$

d : Number of input variables;

N : Number of training samples

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1d} \\ 1 & x_{21} & x_{22} & \dots & x_{2d} \\ \text{---} & \text{---} & \text{---} & \text{---} & \text{---} \\ 1 & x_{n1} & x_{n2} & \dots & x_{nd} \\ \text{---} & \text{---} & \text{---} & \text{---} & \text{---} \\ 1 & x_{N1} & x_{N2} & \dots & x_{Nd} \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \text{---} \\ y_n \\ \text{---} \\ y_N \end{bmatrix}$$

$\mathbf{X}$ is data matrix

Multiple Linear Regression: Testing Phase

- Optimal coefficient vector $\hat{\mathbf{w}}$ is given by

$$\hat{\mathbf{w}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$\hat{\mathbf{w}} = \mathbf{X}^+ \mathbf{y}$$

where $\mathbf{X}^+ = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$ is the pseudo inverse of matrix $\mathbf{X}$

- For any test example $\mathbf{x}$, the predicted value is given by:

$$\hat{y} = f(\mathbf{x}, \hat{\mathbf{w}}) = \hat{\mathbf{w}}^T \mathbf{x} = \sum_{i=0}^d \hat{w}_i x_i$$

- The prediction accuracy is measured in terms of squared error: $E = (\hat{y} - y)^2$
- Let N_t be the total number of test samples
- The prediction accuracy of regression model is measured in terms of **root mean squared error (RMSE)**:

$$E_{\text{RMSE}} = \sqrt{\frac{1}{N_t} \sum_{n=1}^{N_t} (\hat{y}_n - y_n)^2}$$

Illustration of Multiple Linear Regression: Temperature Prediction - Training

Humidity (x_1)	Pressure (x_2)	Temp (y)
82.19	1036.35	25.47
83.15	1037.60	26.19
85.34	1037.89	25.17
87.69	1036.86	24.30
87.65	1027.83	24.07
95.95	1006.92	21.21
96.17	1006.57	23.49
98.59	1009.42	21.79
88.33	991.65	25.09
90.43	1009.66	25.39
94.54	1009.27	23.89
99.00	1009.80	22.51
98.00	1009.90	22.90
99.00	996.29	21.72
98.97	800.00	23.18

$$\hat{\mathbf{w}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$\hat{\mathbf{w}} = [51.3 \quad -0.2386 \quad -0.0055]^T$$

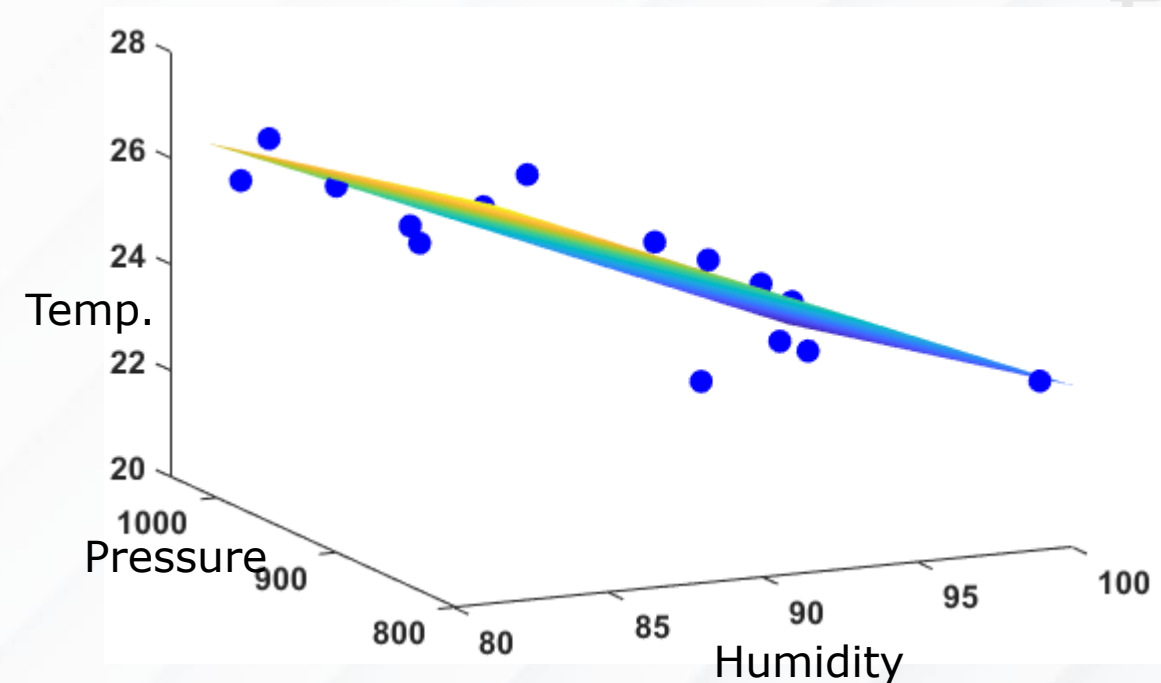


Illustration of Multiple Linear Regression: Temperature Prediction - Test

$$\hat{\mathbf{w}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$\hat{\mathbf{w}} = [51.3 \quad -0.2386 \quad -0.0055]^T$$

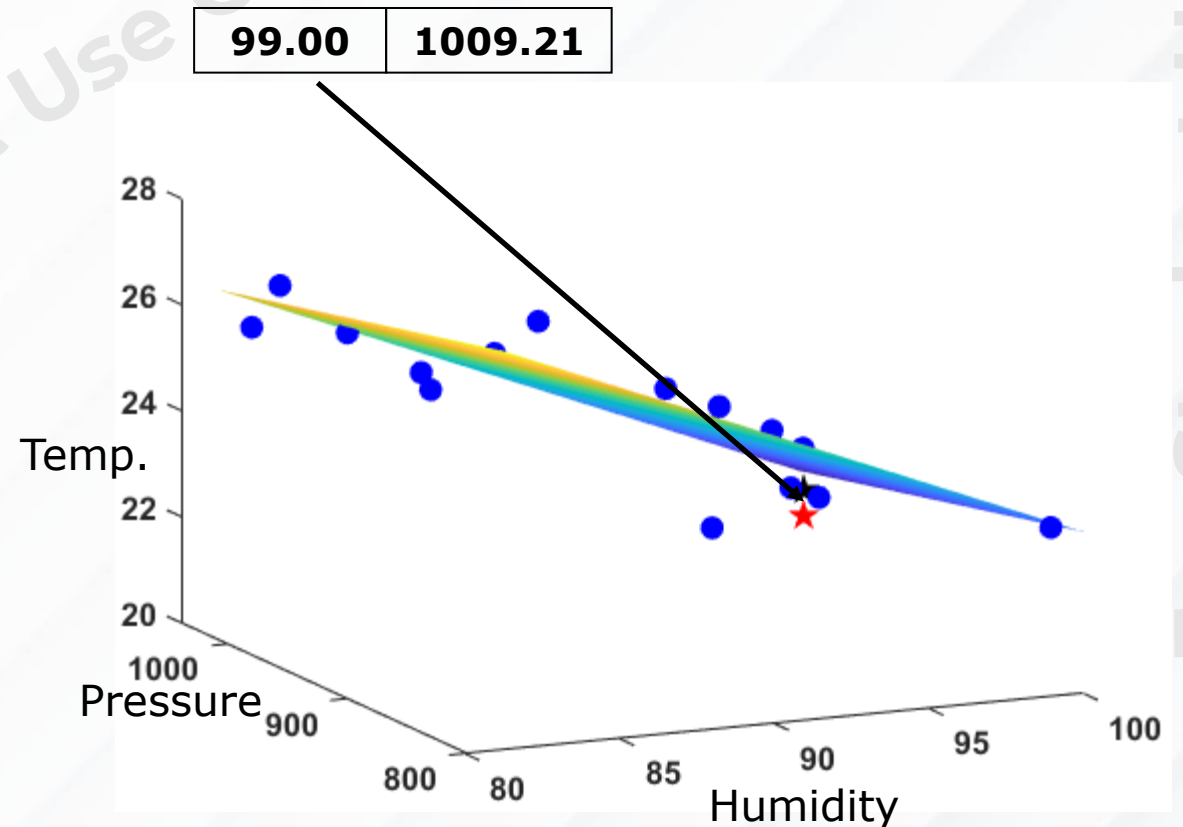
Humidity (x_1)	Pressure (x_2)	Temp (y)
99.00	1009.21	-

$$\hat{y} = f(\mathbf{x}, \hat{\mathbf{w}}) = \hat{\mathbf{w}}^T \mathbf{x}$$

$$\hat{y} = \hat{w}_0 + \hat{w}_1 x_1 + \hat{w}_2 x_2$$

$$\hat{y} = 51.3 + (-0.2386 * 99) + (-0.0055 * 1009.21)$$

- Predicted temperature, $\hat{y}$: 21.72
- Actual temperature, y : 21.24
- Squared error, $E = (\hat{y} - y)^2$: 0.2347



Summary: Linear Regression

- **Regression analysis** is used to model the relationship between **one or more independent** (predictor) variable and **a dependent** (response) variable
- Response is some function of one or more input variables
- **Linear regression**: Response is **linear function** of one or more input variables
 - If the response is linear function of **one input variable**, then it is **simple linear regression** (straight-line fitting)
 - If the response is linear function of **two or more input variables**, then it is **multiple linear regression** (linear surface fitting or hyperplane fitting)

